

Unit: 5

A. Video shooting and Editing:

Definition:

Video shooting is the process of capturing moving images using a video camera or smartphone. It includes planning, framing, lighting, sound recording, and actual filming.

Video editing is the process of selecting, arranging, and modifying video clips using software to create a complete and meaningful final product. It includes cutting scenes, adding music, sound effects, transitions, and text.

Key Points of Video Shooting:

1. Pre-production (Planning):

- Prepare a script or storyboard.
- Select location, cast, and equipment.
- Decide the camera angles and lighting setup.
- Ensure all technical items (battery, memory card, mic) are ready.

2. Camera Techniques:

- Use stable camera positions. A tripod helps to avoid shaky footage.
- Use different shot types:
 - *Wide Shot* (shows full scene)
 - *Medium Shot* (waist up)
 - *Close-Up* (focus on face or detail)
- Maintain proper composition (Rule of Thirds).

3. Lighting:

- Natural light (sunlight) works well, but artificial lights (like LED lights) give more control.
- Avoid harsh shadows or overexposed shots.

4. Sound Recording:

- Use an external microphone for clear audio.
- Avoid noisy backgrounds.

5. B-Roll Footage:

- Take extra shots that can be used to cover cuts or show details. These shots are useful during editing.
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Key Points of Video Editing:

1. Importing and Organizing Footage:

- Transfer all video clips to your computer.
- Use folders and naming systems to organize files.

2. Cutting and Arranging Clips:

- Remove unwanted parts.
- Arrange clips in the right order to match the script or story.
- Keep only the best takes.

3. Adding Transitions:

- Use fade-in, fade-out, or other smooth transitions between clips.

- Don't overuse effects—they should not distract the viewer.

4. **Sound and Music:**

- Adjust audio levels so the voice is clear.
- Add background music or sound effects (but make sure they are not too loud).
- Sync voice and visuals properly.

5. **Text and Titles:**

- Add opening titles, subtitles, and end credits.
- Make sure fonts are readable and placed properly.

6. **Color Correction:**

- Adjust brightness, contrast, and color to make visuals more attractive.
- Maintain consistency in color tone across all clips.

7. **Exporting the Final Video:**

- Choose the right format (like MP4).
- Select suitable resolution (720p or 1080p for good quality).
- Save a backup copy.

Example:

Shooting a 2-minute awareness video on “Clean India Mission”:

- Start with a wide shot of a dirty area.
- Use close-ups to show garbage.
- Record interviews with locals.
- End with a clean-up scene by volunteers.

Editing:

- Cut unwanted parts.
- Add soft background music and voiceover.
- Insert subtitles and a message at the end: *“Keep India Clean – It starts with you.”*

Conclusion:

Video shooting and editing are two important steps in producing visual content. **Shooting** requires planning, good camera work, and clear sound. **Editing** shapes the final video by arranging scenes, adding effects, and improving sound and visuals. Together, they create powerful stories that can educate, entertain, or inspire an audience. Good video work needs creativity, technical skill, and attention to detail.

B. Principles of Video Editing – Linear and non-linear editing, jump cut, cutaway, imaginary line and post-production:

Definition:

Video editing is the process of selecting and combining video shots into a complete story or message. It includes cutting, rearranging, adding effects, sound, and music. There are several principles and techniques editors follow to make the video smooth, understandable, and attractive.

1. Linear Editing

Definition:

Linear editing is the old traditional method where the editor edits the video in a straight sequence — from beginning to end — using tape recorders or VCRs.

Key Points:

- Edits must be made in the exact order.
- Very difficult to make changes once done.
- Time-consuming and less flexible.
- Used before digital technology.

Example:

In the 1980s and early 90s, TV editors used linear editing to cut and arrange scenes using VHS tapes.

2. Non-Linear Editing (NLE)

Definition:

Non-linear editing is the modern digital method where the editor can access any part of the video at any time using editing software.

Key Points:

- Highly flexible and fast.
- Edits can be made in any order.
- Easy to rearrange scenes, add effects, music, and text.
- Popular software: Adobe Premiere Pro, Final Cut Pro, DaVinci Resolve.

Example:

Today's YouTubers use non-linear editing software to mix clips, add titles, background music, and export high-quality videos.

3. Jump Cut

Definition:

A jump cut is an abrupt change in the video where the subject appears to "jump" forward in time, usually within the same frame.

Key Points:

- Often used to show time passing.
- Can look unnatural if not used carefully.
- Sometimes used creatively for fast-paced or comic effects.

Example:

In a vlog, when a person is talking and some parts are removed quickly, it creates jump cuts to keep the video fast and interesting.

4. Cutaway

Definition:

A cutaway is a shot inserted between two main shots to show something related but different.

Key Points:

- Used to hide a jump cut.
- Adds more detail or emotion to the story.
- Helps improve flow and keep viewers interested.

Example:

In a documentary, if someone is talking about a flood, the editor may add cutaway shots of water damage, rescue operations, or maps.

5. Imaginary Line (180-Degree Rule)

Definition:

The imaginary line is an invisible line drawn between two characters or points of action. According to the 180-degree rule, the camera must stay on one side of this line to maintain screen direction.

Key Points:

- Keeps the scene visually clear for viewers.
- Breaking this rule can confuse the audience.
- Used mostly in conversations or action scenes.

Example:

In a film conversation scene, if one person is on the left and the other is on the right, they should always face each other in the same direction. Moving the camera across the line will flip their positions and confuse the viewer.

6. Post-Production

Definition:

Post-production is the final stage after shooting is complete. It includes editing video and sound, adding effects, color correction, titles, and exporting the final product.

Key Points:

- Involves cutting footage, fixing audio, adding transitions, and music.
- Special effects (VFX), graphics, and subtitles are included.
- Color grading and sound mixing are also done here.
- Requires software and skilled editors.

Example:

After a short film is shot, the post-production team edits the scenes, adds background music, syncs sound, adjusts lighting, and prepares the final version for YouTube or film festivals.

Conclusion:

Video editing is both a technical and creative process. The **principles of editing** — including linear vs. non-linear editing, **jump cuts**, **cutaways**, and the **imaginary line rule** — help editors tell a story clearly and smoothly. **Post-production** is the final polish that turns raw footage into a professional video. A good editor understands these principles to create engaging and meaningful content that connects with the audience.

C. TV News – types of news stories (Dry, story with graphics, AVD, story with voice over, news capsule, phone-in, hot switching:

Definition:

TV news includes different styles of presenting news stories on television. Each type has a different format and purpose, depending on the situation, content, and visuals available. News editors and producers choose the suitable type to make the news clear, interesting, and easy to understand for viewers.

Types of TV News Stories:

1. Dry Story

Definition:

A dry story is a basic news story where only the newsreader reads the script. There are no visuals, graphics, or video clips.

Key Points:

- Only anchor is seen on screen.
- No supporting visuals or video footage.
- Often used for short updates or breaking news without video.

Example:

“The Finance Minister has announced a new tax policy today.” (No visuals shown)

2. Story with Graphics

Definition:

This story includes text, charts, photos, maps, or infographics shown on screen while the anchor reads or explains the news.

Key Points:

- Helps explain complex topics (e.g. budget, elections, statistics).
- Makes the story more attractive and easy to understand.
- Used when real video is not available.

Example:

During election news, a pie chart showing vote percentages is shown while the anchor explains the results.

3. AVD (Audio Visual Description)

Definition:

AVD means a complete news story with both video footage and background narration (voice-over).

Key Points:

- Includes visuals like location shots, events, or interviews.
- Narrator explains the story over the video.
- Most common format used in TV news.

Example:

A report on flood damage showing waterlogged streets, damaged houses, and people being rescued — all while the voice-over explains the situation.

4. Story with Voice Over

Definition:

This format has pre-recorded video footage playing while the anchor or a reporter gives an explanation.

Key Points:

- Short and direct format.
- Used for regular news items.
- Voice matches the video scenes for clarity.

Example:

For a school reopening news, footage of children entering the classroom plays while the voice says, “Schools in Delhi reopened today after two months...”

5. News Capsule

Definition:

A news capsule is a short, compact news program that includes multiple top stories, usually in brief.

Key Points:

- Fast-paced summary of main news.
- Usually 5–10 minutes long.
- Often shown in the morning or night.

Example:

A morning bulletin shows headlines like: “1) Train accident in Bihar. 2) Weather alert in Bengal. 3) IPL match results...” — each for a few seconds.

6. Phone-in

Definition:

A phone-in story includes a live or recorded phone call with a reporter or expert who gives updates or opinions.

Key Points:

- Used in breaking news or remote locations.
- Reporter is not on video, only voice is heard.
- Anchor usually asks questions from the studio.

Example:

Anchor: “We are now joined on the phone by our reporter in Kashmir. What’s the situation there?”

Reporter (on call): “There is heavy snowfall, and the highway is closed...”

7. Hot Switching

Definition:

Hot switching means quickly changing the broadcast feed from one live source to another without delay or edit.

Key Points:

- Used during live events, breaking news, or multiple reporters covering different places.
- Requires technical coordination in the control room.
- Keeps the audience updated in real-time.

Example:

While covering election results, the channel switches live from the counting center in Delhi to the party office in Kolkata without a break.

Conclusion:

TV news uses various formats like **dry stories**, **graphics**, **voice-overs**, **AVDs**, and **live inputs** to present information effectively. Each type serves a unique role in storytelling — some explain with visuals, others give quick updates or connect with live callers. A good news producer chooses the right format to make news reports clear, reliable, and engaging for television viewers.

D. Script for television:

Definition:

A **television script** is a written document that contains the **spoken words (dialogue)**, **visual instructions (camera shots, graphics)**, and **technical cues** used during a TV program. It guides news anchors, reporters, cameramen, directors, and editors in making a smooth and well-organized television broadcast.

Key Points:

1. Two-Column Format

Most TV scripts use a **two-column format**:

- **Left column:** Video instructions (camera angles, graphics, visuals)
- **Right column:** Audio/Dialogue (what the anchor or reporter says)

VIDEO

AUDIO

Anchor on camera (Medium shot) “Good evening. You’re watching City News...”

Cut to footage of market

“Today, markets in Kolkata saw heavy crowds...”

Show interview clip

“Here’s what a shopkeeper had to say.”

2. Parts of a TV News Script:

1. **Slug:** Short title of the story (used in newsroom)
 - Example: “Flood Update - Bihar”
2. **Intro:**
 - Anchor reads introduction from the studio.
 - Should be short, catchy, and give basic idea.
3. **Body:**
 - Main details of the story.
 - Includes facts, figures, location, and reactions.
4. **AV Instructions** (Audio/Video):
 - Mention if visuals, graphics, or interviews will play.
 - Use terms like VO (Voice Over), SOT (Sound on Tape), GFX (Graphics), etc.
5. **Sign-Off:**
 - Used when reporter ends a field report.
 - Example: “With cameraman Ravi, this is Asgar Ali reporting for Khabar Setu, Malda.”

3. Script Writing Tips:

- Use **short and clear sentences**.
- Write for the **ear** (how it sounds), not like a newspaper.
- Use **simple language**; avoid long or difficult words.
- Include **names, places, and dates** clearly.
- Time the script — TV news is time-limited.

Example of a Simple TV News Script (2-column format):

VIDEO

AUDIO

VIDEO

Anchor on screen (Studio)

Cut to visuals of heavy rain and water-logging

Map of affected areas

SOT – Interview with local resident

Rescue operation visuals

Reporter on location (Live or recorded)

Sign-off shot (Reporter with mic)

Anchor back on screen

AUDIO

“Good evening. This is Khabar Setu. I’m Asgar Ali with today’s top story.”

“Heavy rains have flooded several areas of Malda district today.”

“According to reports, English Bazar and Kaliachak are worst hit.”

“We haven’t seen such rain in years,” said one resident.

“NDRF teams are working to rescue people in low-lying areas.”

“The situation remains critical here. Water has entered many homes.”

“With cameraman Soumen, this is Asgar Ali for Khabar Setu, Malda.”

“Thank you, Asgar. We’ll keep tracking this story.”

Conclusion:

A **television script** is the backbone of a TV program. It ensures clear communication between the **anchor**, **reporter**, **director**, and **technical team**. With proper use of visuals, sounds, and spoken words, the script helps deliver news in a smooth, professional, and viewer-friendly way. Whether it is a **live bulletin**, **recorded report**, or a **feature story**, a well-written script makes television broadcasting successful.

E. TV programmes and documentaries:

Definition:

- A **TV Programme** is any content that is broadcast on television for entertainment, education, information, or awareness.
- A **Documentary** is a non-fiction television programme that presents real-life events, people, or issues using facts, research, interviews, and visuals.

1. TV Programmes:

Definition:

TV programmes are various types of content shown on television that can include news, drama, reality shows, quiz shows, talk shows, and educational programmes.

Key Points:

- TV programmes can be **fiction** (like serials and dramas) or **non-fiction** (like news and talk shows).
- Different genres serve different purposes: some to entertain, some to inform, and some to educate.
- They are usually made for regular broadcasting with planned time slots.

Examples of TV Programmes:

- **News Bulletin** (Example: DD News at 9 PM)
- **Reality Show** (Example: Kaun Banega Crorepati)
- **Talk Show** (Example: Aap Ki Adalat)
- **Comedy Show** (Example: The Kapil Sharma Show)
- **Daily Soap / Serial** (Example: Anupamaa)
- **Children's Programme** (Example: Balveer)

2. Documentaries:

Definition:

A documentary is a detailed and factual video story based on **real events, social issues, historical topics, or people's lives**. It is made after research and usually contains interviews, real footage, narration, and sometimes archive material.

Key Points:

- Based on **truth and research**.
- Focus on **awareness, education, and social change**.
- Include visuals, interviews, voice-over, facts, statistics, and location shots.
- Usually longer than news reports and require more planning and production time.

Types of Documentaries:

1. **Historical** – Covers past events (e.g. India's Freedom Struggle).
2. **Biographical** – About a person's life (e.g. Life of Dr. APJ Abdul Kalam).
3. **Investigative** – In-depth research on serious topics (e.g. corruption or scams).
4. **Environmental** – Focus on nature and issues (e.g. climate change, deforestation).
5. **Social Issue-based** – Explains current problems (e.g. child marriage, poverty).

Example of Documentary Titles:

- *India: The Story of Us* (History)
- *Plastic Ocean* (Environment)
- *Daughters of Destiny* (Social issue – education of girls)
- *BBC's Earth Series* (Nature and Wildlife)

Difference Between TV Programme and Documentary:

Point	TV Programme	Documentary
Type	Fiction and Non-Fiction	Always Non-Fiction
Purpose	Entertainment, information, fun	Education, awareness, social message
Content Style	Can be drama, reality, game, etc.	Real events, facts, research-based
Duration	Usually short (30 min–1 hour), regular	Can be short or long, usually one-time
Examples	News, Serial, Quiz Shows	Earth Report, India Untold, Women Empowered

Conclusion:

TV programmes and **documentaries** both play an important role in the television industry. While TV programmes focus on entertainment, time-based content, or public engagement, **documentaries are powerful tools** used to **educate, inform, and inspire change**. A good television media house should balance both — creative shows for mass appeal and truthful documentaries for meaningful impact.

F. Development of TV in India – SITE, Cable TV, DTH, IPTV, HITS:

Definition:

The development of television in India refers to the historical and technological growth of the television broadcasting system, from its government-controlled beginnings to private and digital systems like DTH, IPTV, and HITS. It represents how India expanded mass communication through television for education, entertainment, and national integration.

Key Stages with Authentic Data:

1. SITE (Satellite Instructional Television Experiment) – 1975–1976

- **Full form:** Satellite Instructional Television Experiment.
- **Organised by:** Indian Space Research Organisation (ISRO) in collaboration with NASA.
- **Experiment period:** 1 August 1975 – 31 July 1976.
- **Satellite used:** ATS-6 (Application Technology Satellite - 6) provided by NASA.
- **Budget:** Around ₹10 crore (funded by ISRO and government).
- **Coverage:** Over **2400 villages in 6 states** – Bihar, Orissa, Madhya Pradesh, Andhra Pradesh, Karnataka, and Rajasthan.
- **Program themes:** Agriculture, health, family planning, primary education, hygiene.
- **Languages used:** Hindi, Oriya, Kannada, Telugu.
- **Daily broadcast:** 4 hours a day for one year.
- **Objective:** Educate rural masses using satellite technology.

Significance:

- Marked India's **first step in satellite-based communication**.
- Recognised globally as one of the most successful experiments in development communication.
- Proved that **TV can be used for social upliftment**, not just entertainment.

Quote from Dr. Vikram Sarabhai: “Television can be a very powerful tool for national development.”

2. Cable TV – Introduced in India in 1984, Growth in 1990s

- **First appearance:** Around **1984** in cities like Mumbai, Delhi, Kolkata, and Chennai.
- **Massive growth after: 1991 Gulf War**, when CNN showed live war coverage – introduced foreign channels to Indian homes.
- **Legal framework:** Cable Television Networks (Regulation) Act, 1995.
- **Early private channels:** Zee TV (1992), Star TV (1991 in Asia), Sun TV (1993), Sony (1995).
- **Viewership expansion:** From 41 TV sets in Delhi in 1959 to **over 410 million TV viewers in India by 2023**.
- **Cable networks offered:** 50–100+ channels via coaxial cable to homes.

Significance:

- Brought **private sector and commercialisation** in Indian television.
- Provided **regional content**, movies, serials, 24x7 news.
- Increased **TV penetration** in urban and semi-urban areas.

According to **TRAI (2022 report)**, 1039 Multi-System Operators (MSOs) are registered to operate Cable TV in India.

3. DTH (Direct-To-Home) – Started in 2003

- **First Indian DTH service:** Dish TV, launched on 2 October 2003 by Zee Group.
- **Other DTH providers:**
 - Tata Sky (now Tata Play) – 2006
 - Airtel Digital TV – 2008
 - Sun Direct – 2007
 - Videocon d2h – 2009
 - DD Free Dish (Free DTH by Doordarshan) – Launched in 2004
- **Technology used:** MPEG-2, later MPEG-4 digital compression.
- **Features:**
 - 200–600+ channels, HD quality, radio, interactive services.
 - Set-top box with dish antenna required.
 - Pay-per-view movie services, kids zone, sports packages, etc.
- **As of TRAI Report (2023):**
 - India had **over 65 million active DTH subscribers** (excluding DD Free Dish).

Significance:

- Enabled access to **remote and rural areas**.
- **Improved quality** of audio-visual transmission.
- Allowed viewers to **customise channel packs**.

DD Free Dish is the world's **only free DTH platform**, with over **50 million users**, mainly in rural India.

4. IPTV (Internet Protocol Television) – Around 2008 Onwards

- **Definition:** Transmission of TV content through broadband internet.
- **Major IPTV services launched:**
 - MTNL IPTV in Delhi and Mumbai (2008)
 - BSNL IPTV (2009)
 - Airtel Xstream, Jio TV, ACT Stream TV, etc.
- **Working:** Content is streamed over IP networks to smart TVs, mobiles, or set-top boxes.
- **Features:**
 - On-demand services (VoD), live TV, catch-up TV.
 - Pause and rewind live shows.
 - Interactive features (polling, voting, etc.)
- **Required:** High-speed broadband connection.

Significance:

- Useful in **urban and tech-savvy households**.
- Facilitated **convergence** of TV and internet.
- Gave rise to OTT platforms (like Hotstar, Netflix, Amazon Prime).

According to a **KPMG report (2022)**, IPTV and OTT streaming account for **over 50% of digital media consumption** in India.

5. HITS (Headend In The Sky) – Introduced in 2010

- **Definition:** Centralized satellite-based digital distribution system for cable TV operators.
- **How it works:**
 - HITS operator collects channels at the uplink centre.

- Channels are beamed to a satellite (like GSAT-10).
- Local cable operators downlink and distribute to homes.
- **Popular HITS platform: NXT Digital** (by Hinduja Group).
- **Approved by:** Ministry of Information and Broadcasting.
- **Purpose:** Support small cable operators to digitize easily.
- **TRAI Guidelines:** Ensure every household gets digital TV services.

Significance:

- Helped in **cable digitization phase** under Government's plan (2012–2017).
- Brought **uniform quality** to tier-2, tier-3 cities and villages.
- Strengthened **last-mile connectivity** for cable networks.

Example (Rural India Journey):

In **Kaliachak (Malda district)**, during the 1975 SITE program, villagers saw health awareness videos. In the 1990s, they enjoyed Zee TV through local cable. By 2010, Tata Sky brought DTH services with Bengali channels. Today, students stream online classes through **IPTV and OTT apps** like JioTV, and cable operators use HITS to deliver HD channels.

Conclusion:

The development of television in India shows the transition from **a single government broadcaster (Doordarshan)** to a **vibrant, multi-platform system**. From **SITE's rural education** to **DTH's digital access**, and now **IPTV and HITS enabling smart and uniform services**, the journey reflects **technological innovation, policy support, and social change**. Television has become a powerful tool in shaping India's **information society** and promoting **education, culture, and democracy**.

G. Formats of TV signals – SDTV, HDTV:

Definition:

TV signal formats refer to the different resolutions and quality levels at which television content is recorded, transmitted, and displayed. These formats affect the **picture clarity, aspect ratio, frame rate**, and overall **viewing experience**. The two most common types are:

- **SDTV (Standard Definition Television)**
- **HDTV (High Definition Television)**

1. SDTV – Standard Definition Television

Meaning:

SDTV is the **basic digital format** of television broadcasting, with lower picture resolution compared to high-definition formats. It was the **standard format before the arrival of HD** and is still used widely in many regions.

Key Features:

- **Resolution:**
 - In PAL (used in India): **720 × 576 pixels**
 - In NTSC (used in USA): **720 × 480 pixels**
 - **Aspect Ratio:**
 - Traditionally 4:3 (square-like screen)
 - Some SD channels also broadcast in 16:9 (widescreen)
 - **Frame Rate:**
 - 25 frames per second (PAL)
 - 30 frames per second (NTSC)
 - **Bandwidth Required:** Less than HDTV (suitable for analog or low-speed digital systems)
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Advantages:

- Needs **less bandwidth** for transmission.
 - **Compatible** with older television sets.
 - Still used for **regional and free-to-air channels** in rural areas.
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Limitations:

- **Lower picture clarity.**
 - Poor performance on modern big-screen TVs.
 - Less detailed images and text.
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Example:

- **DD National (SD):** Available free on **DD Free Dish**.
 - **Older channels like Zee TV, Sony** initially broadcast in SD format.
 - Many regional cable operators still provide SDTV in rural India.
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2. HDTV – High Definition Television

Meaning:

HDTV is a **digital television format** that provides much **higher resolution** and better picture quality compared to SDTV. It is designed for **modern digital TVs**, offering **clear, sharp, and lifelike images**.

Key Features:

- **Resolution Options:**
 - **HD Ready:** 1280 × 720 pixels (720p)
 - **Full HD:** 1920 × 1080 pixels (1080i or 1080p)
- **Aspect Ratio:** 16:9 (widescreen format)
- **Frame Rate:** 24, 30, or 60 frames per second
- **Audio Quality:** Dolby Digital or surround sound (better than SD)

Advantages:

- **Sharper and more detailed images**
- **Ideal for large-screen LED or smart TVs**
- **Better for sports, movies, and high-speed content**
- **Better color depth and sound quality**

Limitations:

- Requires **more bandwidth** and **better internet/cable connection**
- Needs **HD set-top box** or **HD-compatible devices**
- Consumes more data when streamed online

Example:

- **Star Plus HD, Sony HD, Zee Bangla HD** available on DTH platforms like **Tata Play, Airtel Digital TV**.
- Most OTT platforms like **Netflix, Amazon Prime** offer content in Full HD and even higher (4K).
- News channels like **Aaj Tak HD, India Today HD** give crystal-clear visual experience.

Comparison Table:

Feature	SDTV	HDTV
Resolution	720×576 (PAL), 720×480 (NTSC)	1280×720 or 1920×1080
Aspect Ratio	4:3 or 16:9	Always 16:9
Picture Quality	Basic	Very clear and detailed
Sound Quality	Basic mono/stereo	Digital surround sound
Bandwidth Requirement	Low	High
Set-Top Box Needed	Normal	HD Set-top Box required

Conclusion:

The evolution from **SDTV to HDTV** shows how television technology has improved over time to give viewers a **better and more immersive experience**. SDTV is still used for basic broadcasting in many regions, especially rural areas, while HDTV has become the **standard format in urban India**. With the rise of **smart TVs and digital platforms**, HDTV has set a **new quality benchmark** for visual entertainment.